

EEAIC - BONUS Exercise

Alessandro Vegni & Giulio Guerrieri

The figures presented below show the layout and schematic views of the designed amplifier. The original file is located in the project directory:

`/home/eeaic_15fs26/assign10/dfii/BONUS/inv_cascode_amp`

The trickiest part of this layout was the instantiation of the M_{4a} , M_{4b} and M_{3a} , M_{3b} cascode pairs, as they do not share a common drain or source. The solution applied in this layout was to insert dummy transistors between the A and B branches, therefore preserving the interdigitated layout pattern. This resulted in a slightly more complex schematic (*Figure 3*) regarding the dummy devices, as they have different drain and source potentials. Nevertheless, their gates are always tied to the maximum or minimum potential (for PMOS and NMOS, respectively), which guarantees that the devices remain off. If two dummy transistors had been placed at every transition between A and B, it would have yielded an even better layout isolation, but it would have required a larger total number of dummy transistors compared to our current layout.

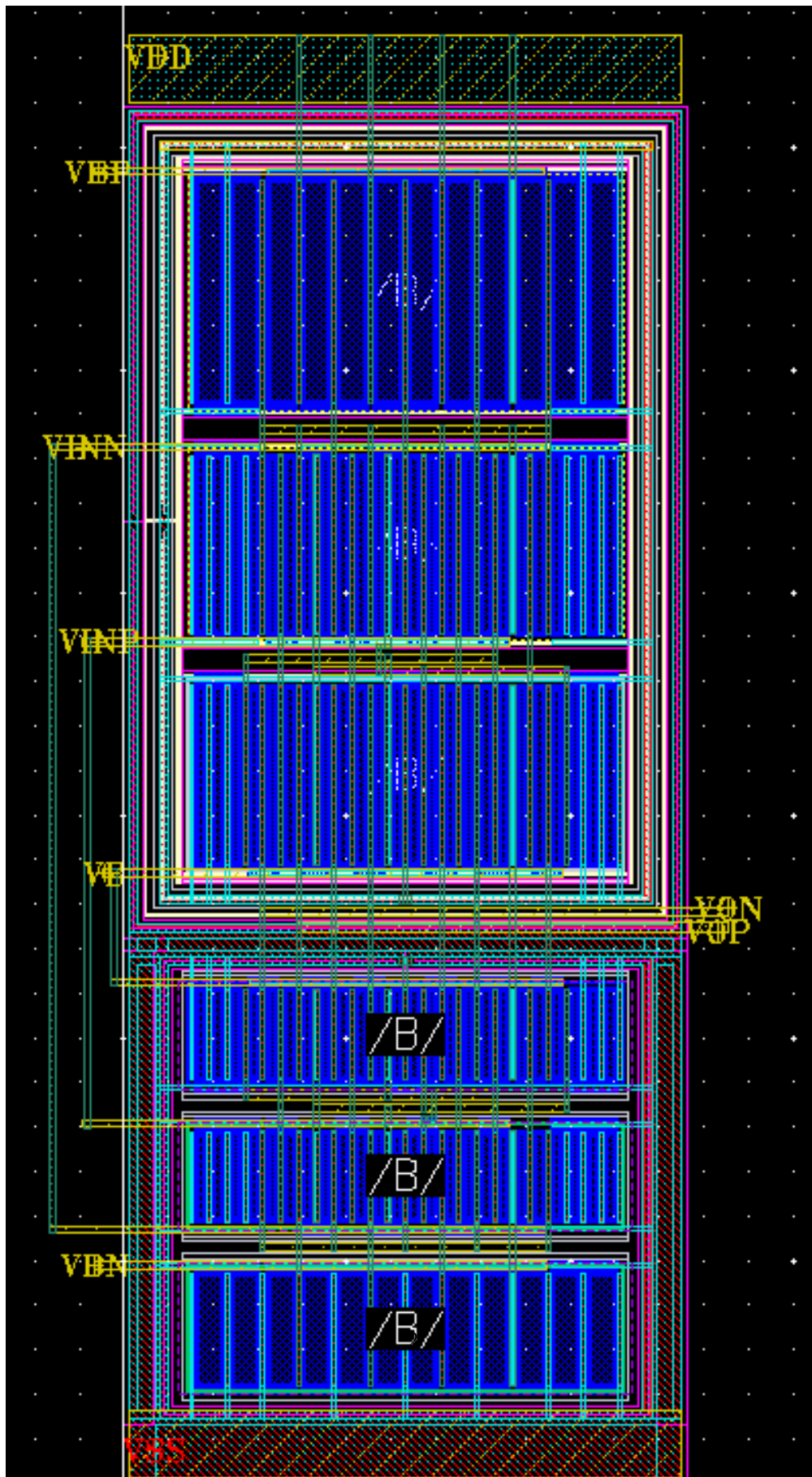


Figure 1: Total Layout

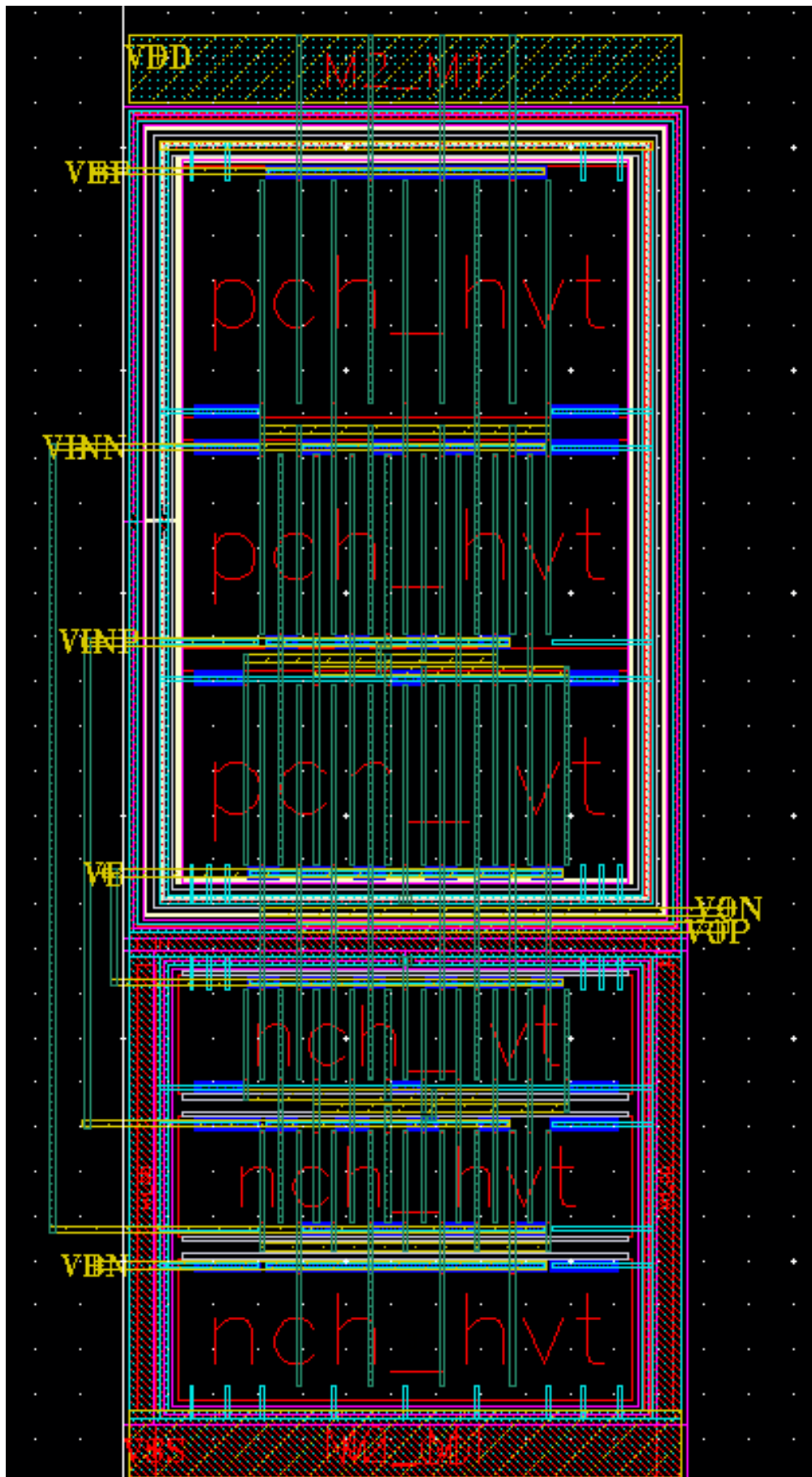


Figure 2: Total Layout with Transistor Types

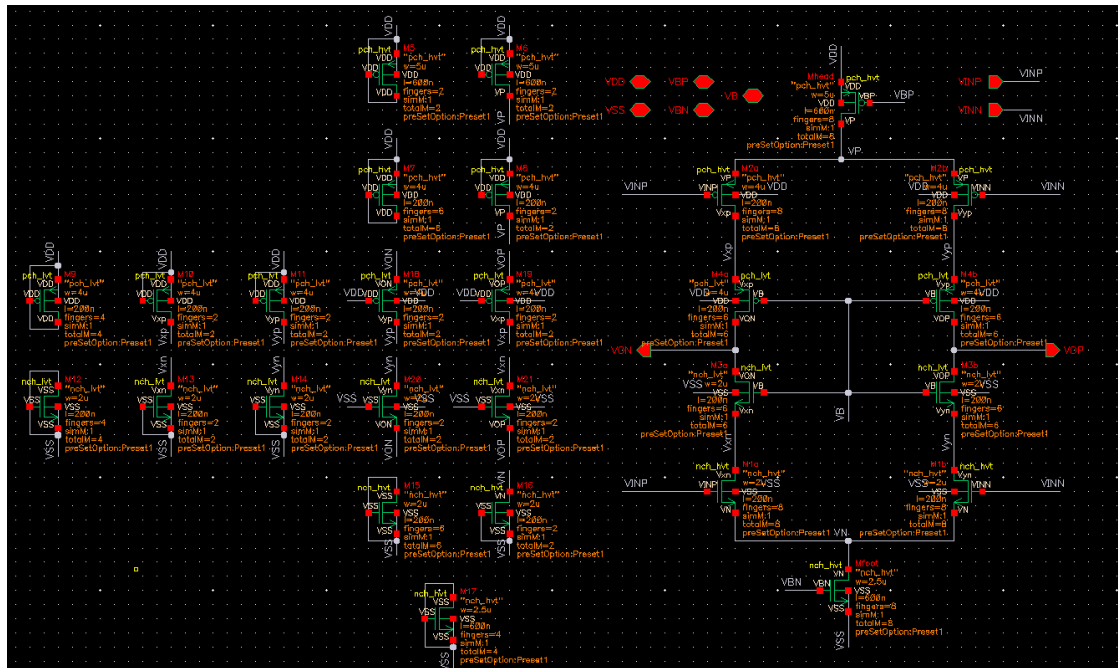


Figure 3: Amplifier Schematic

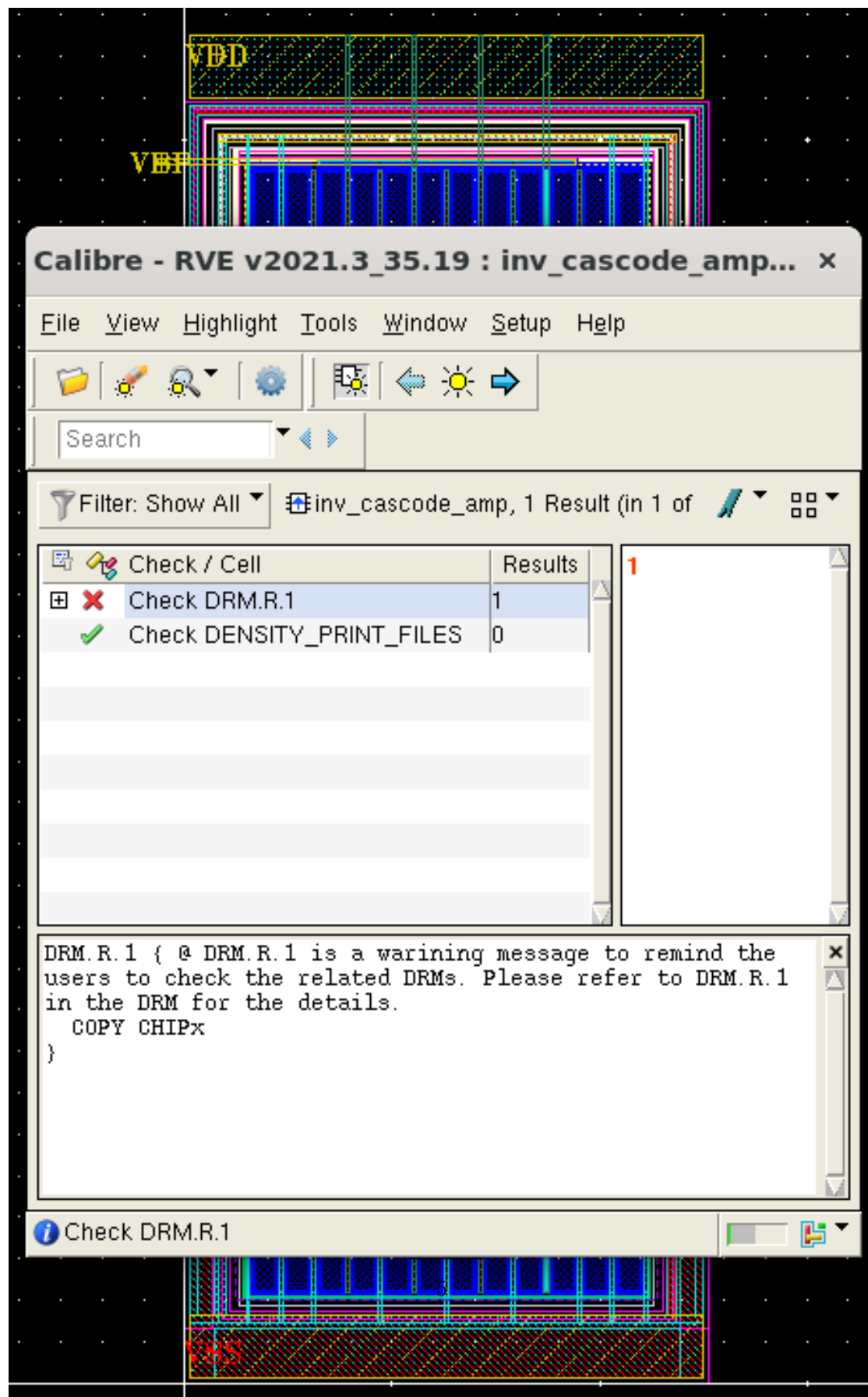


Figure 4: DRC

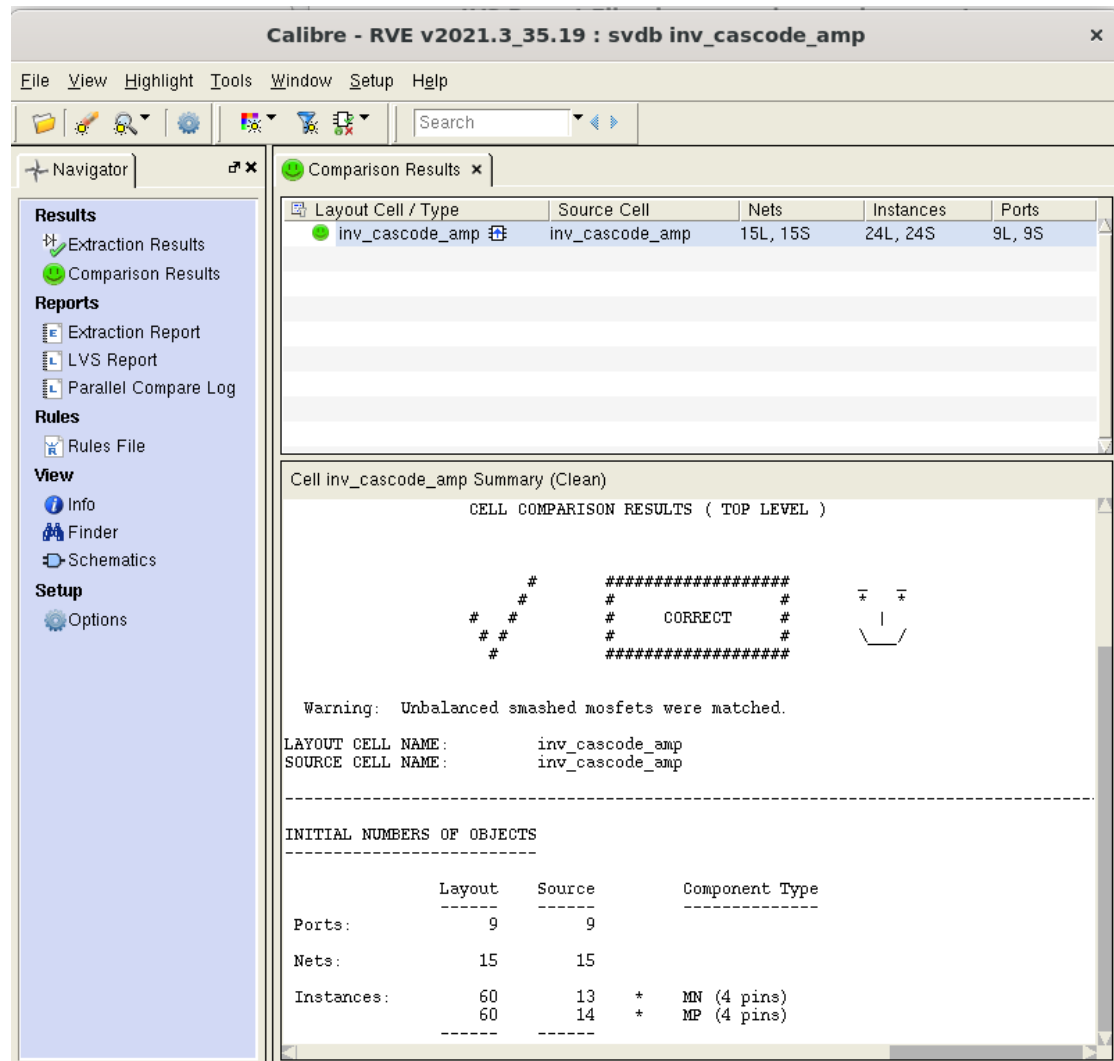


Figure 5: LVS

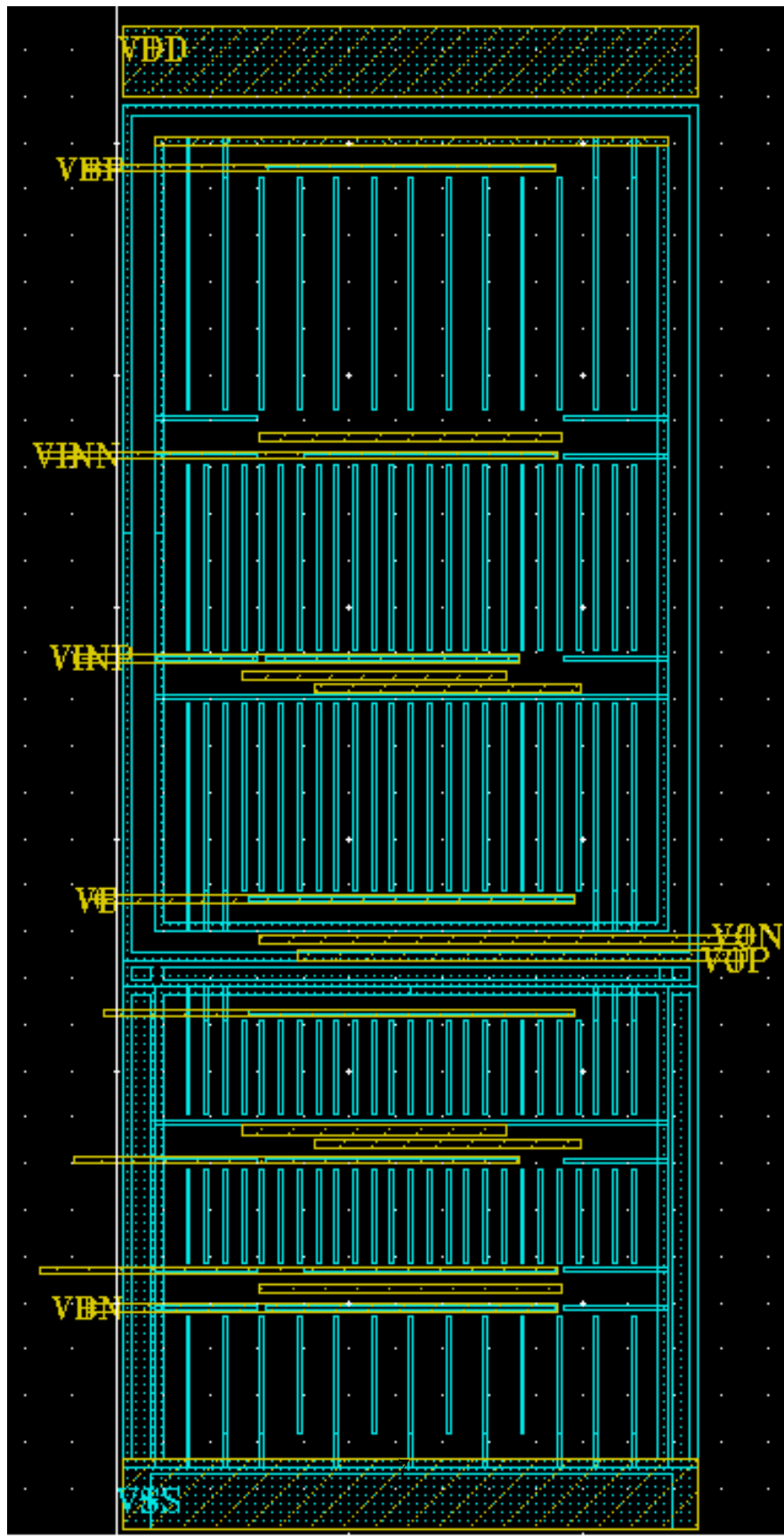


Figure 6: M1 and M2 Layers

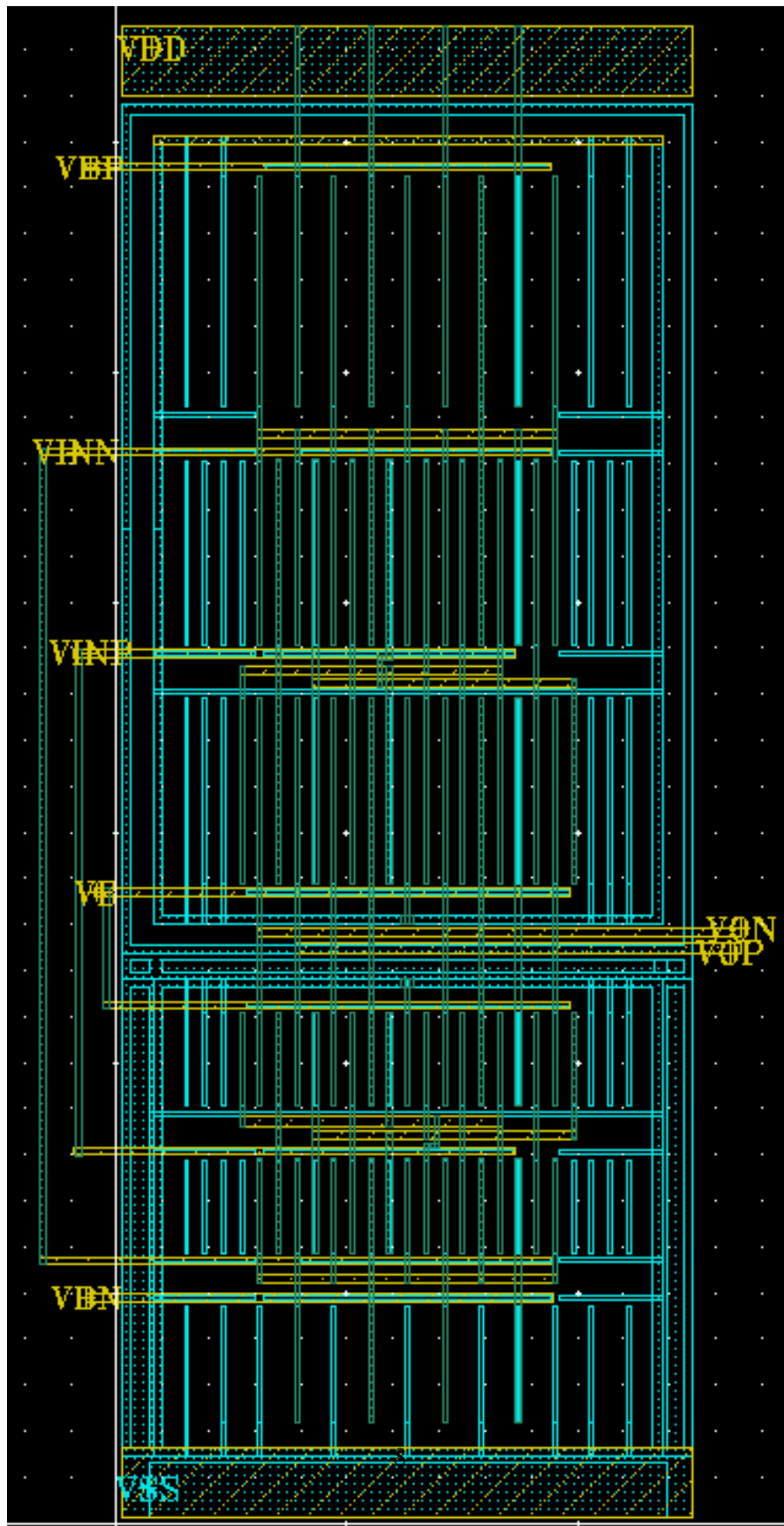


Figure 7: M1, M2, and M3 Layers

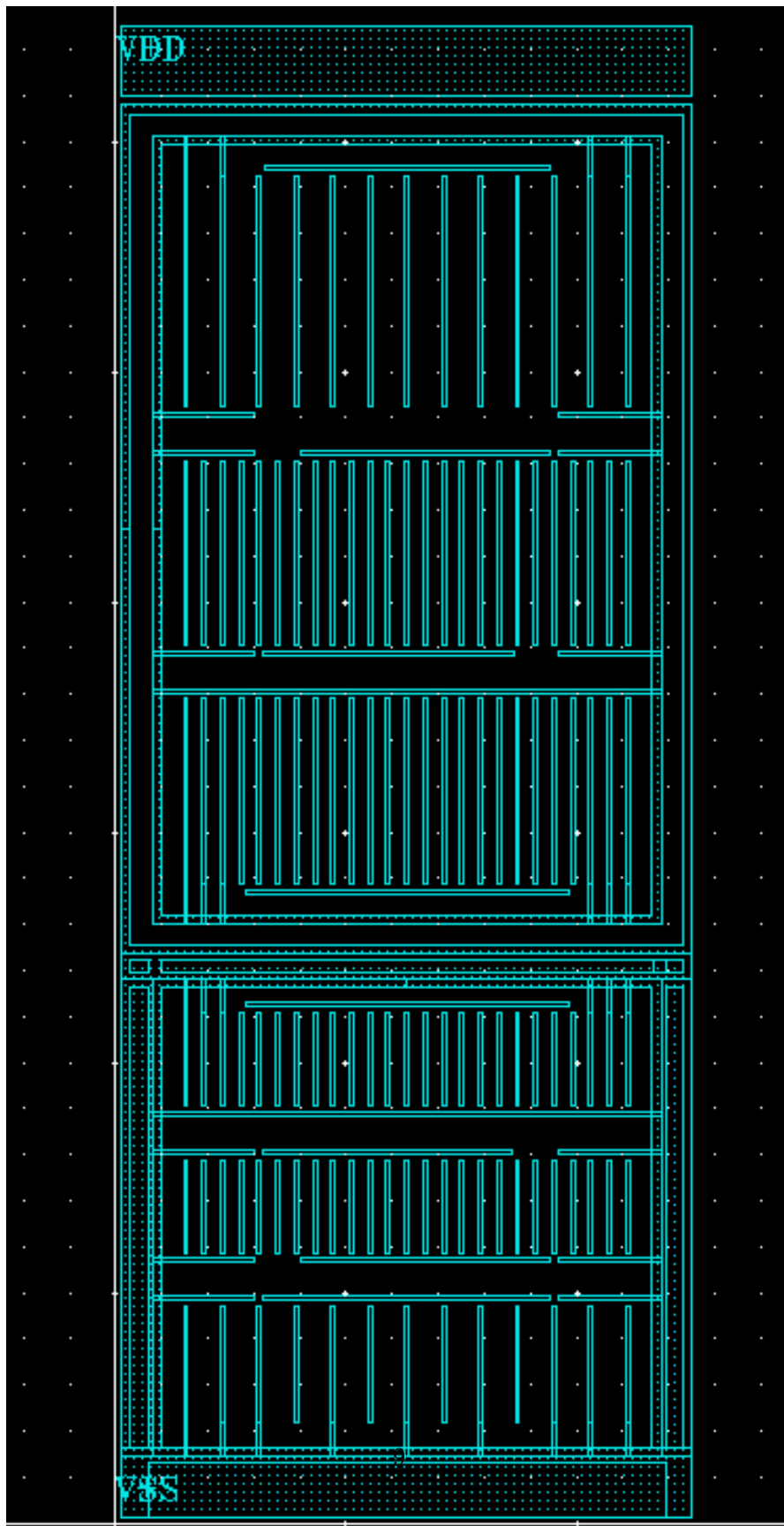


Figure 8: M1 Layer Only

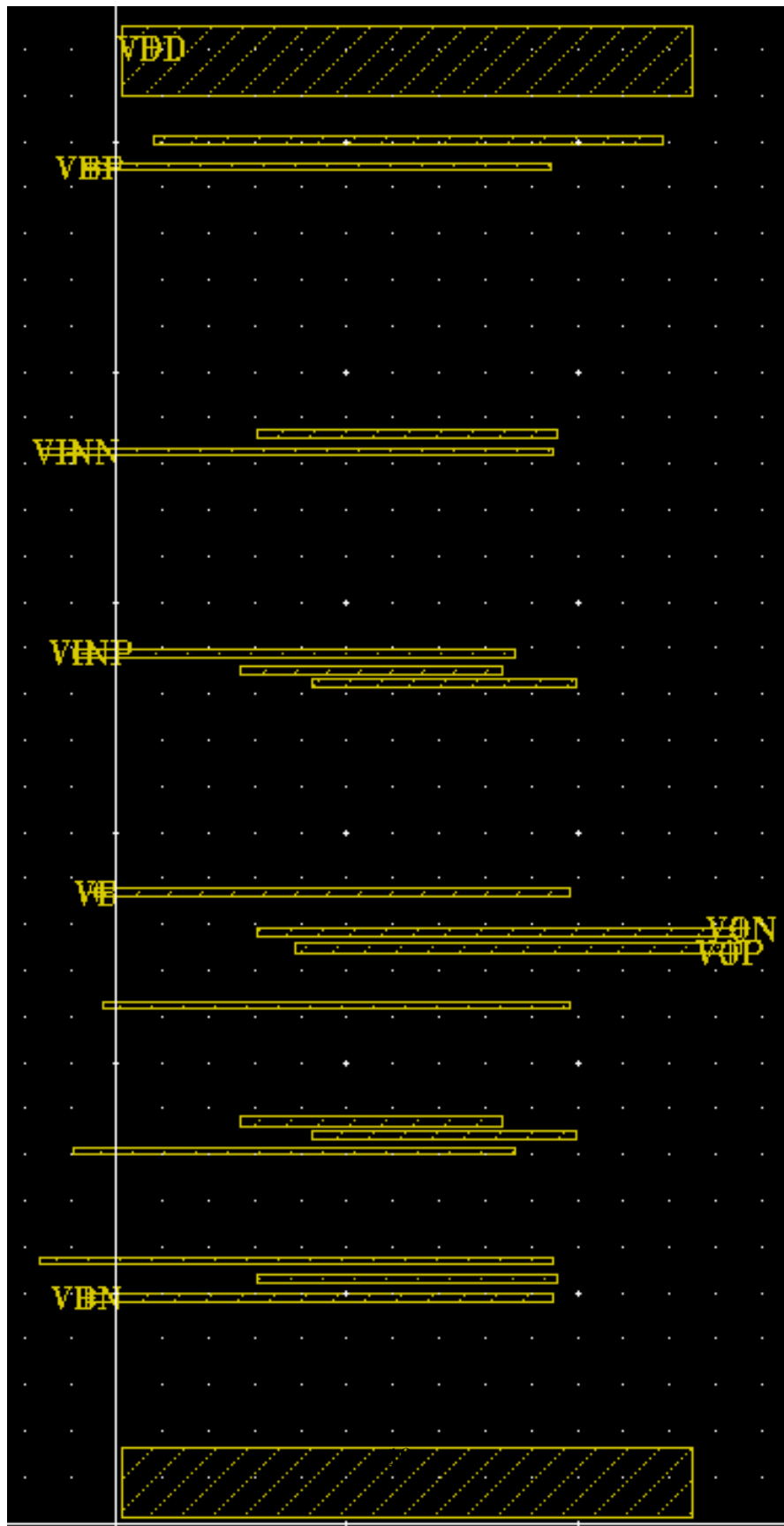


Figure 9: M2 Layer Only

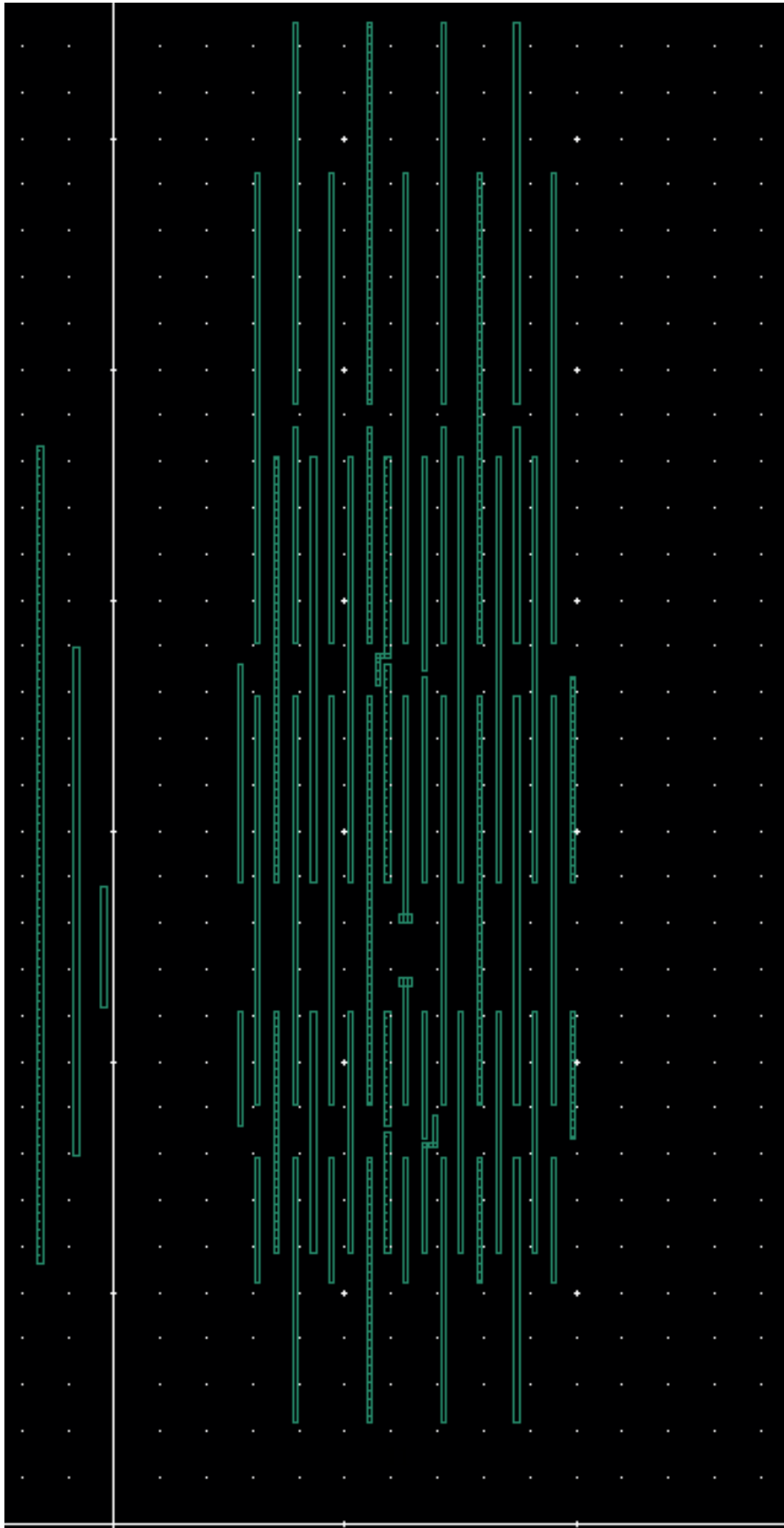


Figure 10: M3 Layer Only